

YINGAO (ELAINE) YAO

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<https://elaineyao.github.io/>

RESEARCH INTERESTS

My research interests are at the intersection of Software Engineering and Machine Learning (ML). I am particularly interested in developing software testing techniques (e.g., fuzzing) to improve the security, reliability, and correctness of ML systems.

EDUCATION

Cornell University

September 2024 - Present

Ph.D. in Computer Science.

Advisor: Dr. Saikat Dutta.

University of British Columbia

September 2021 - December 2023

MASc. Electrical and Computer Engineering.

Advisor: Dr. Karthik Pattabiraman.

University of Electronic Science and Technology of China

September 2017 - June 2021

B.E. Electrical and Communication Engineering

GPA: 3.96/4.0

PUBLICATIONS

Automated Assertion Generation and Regression Testing for Machine Learning Notebooks

ASE 2026

Yingao (Elaine) Yao, Vedant Nimje, Varun Viswanath, and Saikat Dutta.

IEEE/ACM International Conference on Automated Software Engineering (ASE).

🏆 **ACM SIGSOFT Distinguished Paper Award**

ICS-Sniper: A Targeted Blackhole Attack on Encrypted ICS Traffic

NDSS 2027

Gargi Mitra, Chanyuan Liu, Pritam Dash, Yingao Yao, Alain Zhiyanov, Aastha Mehta, Karthik Pattabiraman.

Network and Distributed System Security (NDSS).

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms

DSN 2023

Yingao (Elaine) Yao, Pritam Dash, and Karthik Pattabiraman.

IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms

CCS 2023

Yingao (Elaine) Yao, Pritam Dash, and Karthik Pattabiraman.

ACM SIGSAC Conference on Computer and Communications Security (CCS).

HONORS AND AWARDS

ACM SIGSOFT Distinguished Paper Award

2026

IEEE DSN Student Travel Grant Award

2023

Faculty of Applied Science Graduate Award, University of British Columbia

2022

Outstanding Graduate, Sichuan Province

2021

Outstanding Winner, COMAP Interdisciplinary Contest in Modeling

2020

Thanksgiving Scholarship for Modern Scientists

2020

National Scholarship

2020

National Second Prize in China Undergraduate Mathematical Contest in Modeling

2018

WORKING EXPERIENCE

Security Researcher
Fortinet, Vancouver, Canada.

Jan 2024 - Aug 2024

TEACHING EXPERIENCE

Teaching Assistant for CS 3780/5780: Intro to Machine Learning
Cornell University, Ithaca, NY, USA

Fall 2026

Teaching Assistant for CS 5150: Software Engineering
Cornell University, Ithaca, NY, USA

Spring 2025

Teaching Assistant for CPEN 321: Software Engineering *Spring 2022, Fall 2022, Spring 2023*
University of British Columbia, Vancouver, Canada.

ACADEMIC SERVICE

Artifact Evaluation Program Committee

2024

19th USENIX Symposium on Operating Systems Design and Implementation (OSDI).

European Conference on Object-Oriented Programming (ECOOP).

18th USENIX WOOT Conference on Offensive Technologies (WOOT).

ACM Conference on Computer and Communications Security (CCS).

USENIX Annual Technical Conference (USENIX ATC).

The 54th Annual IEEE/IFIP International Conference on Dependable Systems and Networks (DSN).

Artifact Evaluation Program Committee

2023

The 34th IEEE International Symposium on Software Reliability Engineering (ISSRE).

DEPARTMENT SERVICE

Department of Computer Science, Picnic Czar
Cornell University, Ithaca, NY.

Fall 2025, Spring 2026

MENTORING

David Jovic (BS, University of Belgrade)

Arpan Chakraborty (BS, Adamas University)

Vedant Ramesh Nimje (BS, Veermata Jijabai Technological Institute, Mumbai)

Varun Viswanath (BS, Dwarkadas J Sanghvi College of Engineering)

Parshwa Nandu (BS, Dwarkadas J Sanghvi College of Engineering)

TALKS

Automated Assertion Generation and Regression Testing for Machine Learning Notebooks *ASE 2026*

SwarmFuzz: Discovering GPS Spoofing Attacks in Drone Swarms

DSN 2023

Poster: May the Swarm Be With You: Sensor Spoofing Attacks Against Drone Swarms.

CCS 2023

OPEN-SOURCE CONTRIBUTIONS

SwarmFuzz: <https://github.com/DependableSystemsLab/SwarmFuzz>

NBTest: <https://github.com/seal-research/NBTest>